

Orange Crushing Force Test

Standard Operating Procedure

Objective

- To estimate the maximum amount of force required for an orange crusher to juice an orange

Materials

- 2 Medium-sized oranges at room temperature
- Knife
- Digital Scale
- Paper Towels
- Bowl
- Cleaning Supplies (soap, water)

Procedure

1. Use the knife to slice an orange in half
2. Place one half of the orange in a bowl, cut side down
3. Place the bowl with the orange in it on the digital scale
4. Record initial weight displayed on the scale
5. Zero or calibrate the scale, if necessary, to ensure accurate measurement
6. Gradually apply downward pressure onto the orange using your hand
7. Stop applying force when the orange is flatten
8. Record the maximum force reading shown on the scale
9. Dispose of the crushed orange properly
10. Clean the bowl
11. Repeat steps 1-10 for the next orange
12. Wash hands and clean any surfaces exposed to juice

Means of Measuring Force

- A digital scale was used along with manual pressure to crush the oranges. The scale measured the force by displaying the downward pressure applied, in pounds. As pressure was gradually applied to the orange, the scale registered the increasing weight. The highest reading observed on the scale at the point where the orange became visibly flat was recorded as the maximum crush force.

When You Know have “Finished” the Test

- The test will be considered “finished” when the orange is visibly flattened, juice leaked from the orange, and the maximum weight reading on the digital scale has been reached.

Results and Conclusion

Results

Trial	Peak Scale Reading (lbf)
1	46.62
2	49.39
3	48.27
4	47.38

- The average crushing force was 47.92 pounds-force (lbf). The results of my test indicated that a force of at least 47.92lbf is required to crush half of an orange.

Conclusion

- I am moderately confident that my test setup provided a reasonable estimate of the force required to crush half an orange. One limitation was the fluctuation in readings from the digital scale, which made it difficult to capture an exact maximum value at times. Also, the use of manual pressure introduced a margin of error. The angle at which I pushed down on the orange could have influenced the measurements, though I tried to keep my arm straight. My hand placement mirrored that of the orange crusher I designed. However, my hand placement and inconsistent application of pressure might have affected the measurements. Also, I did not slice the oranges directly in the middle, so that may have influenced the measurements. The ripeness and size of the orange will influence the force needed to crush it in a crusher.